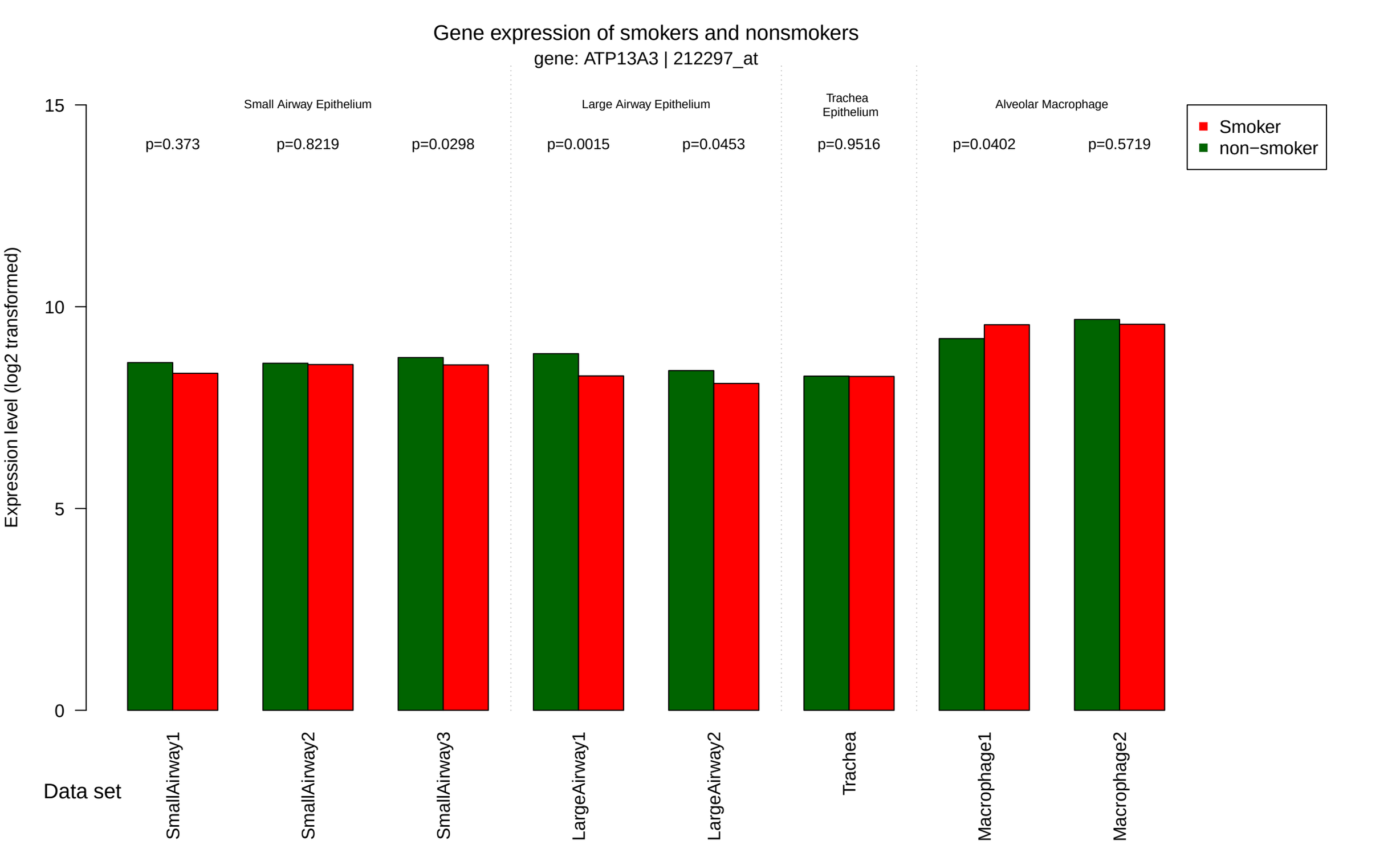
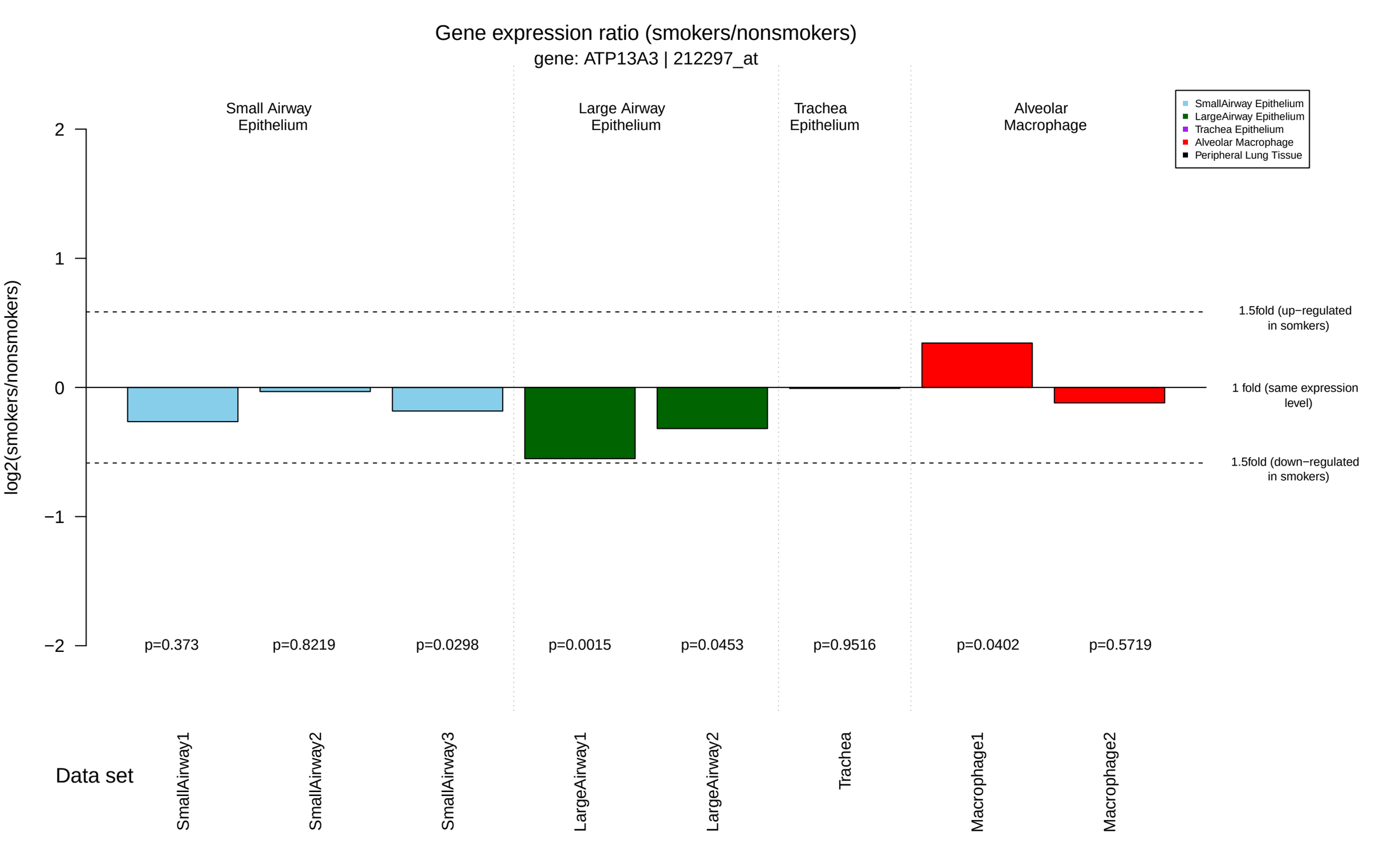
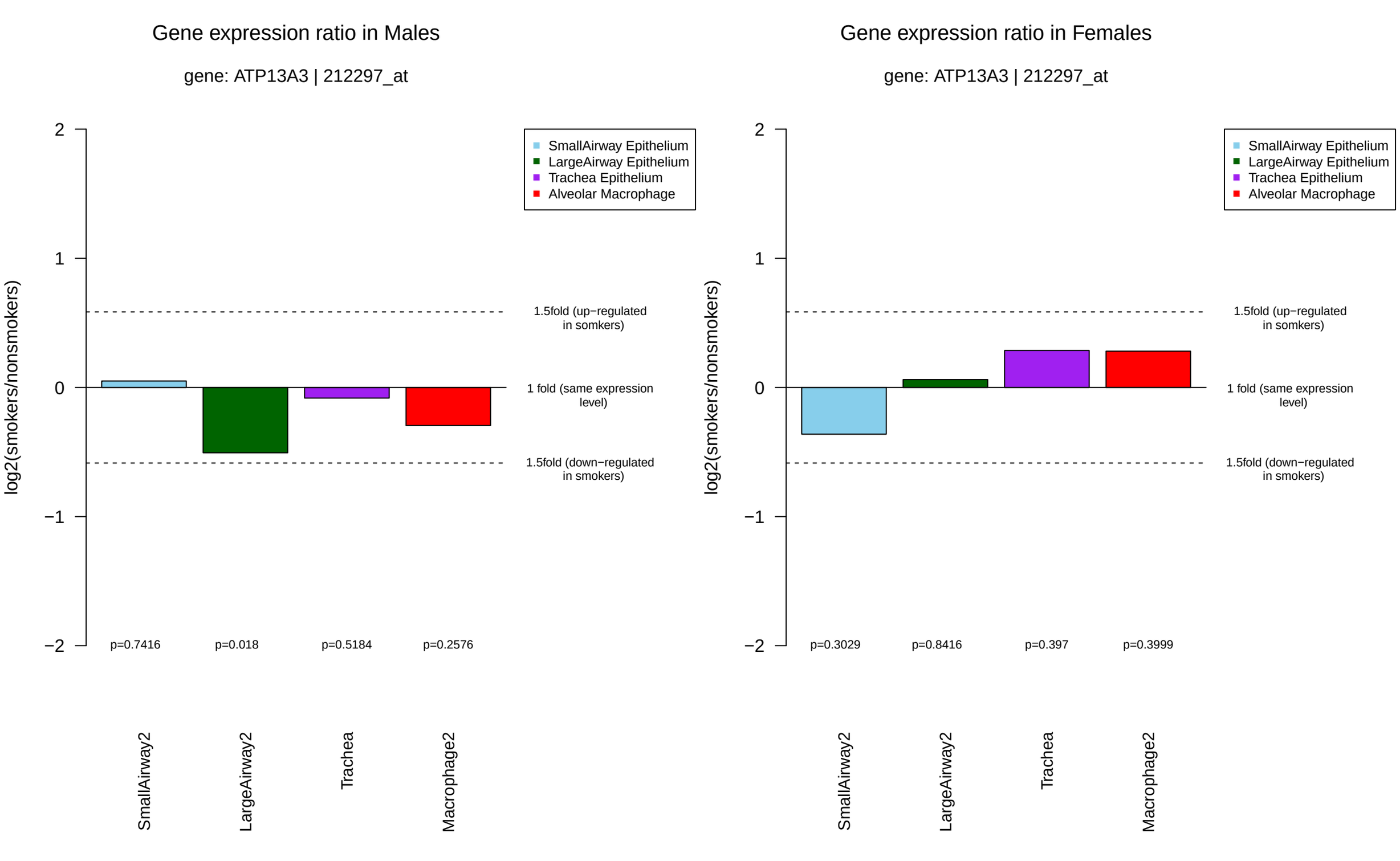
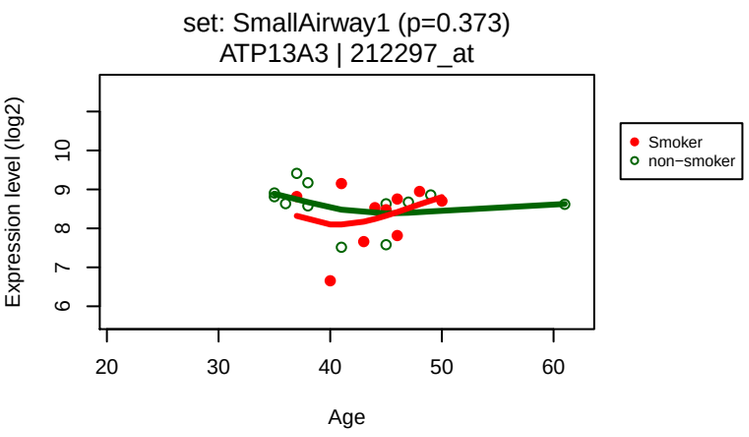
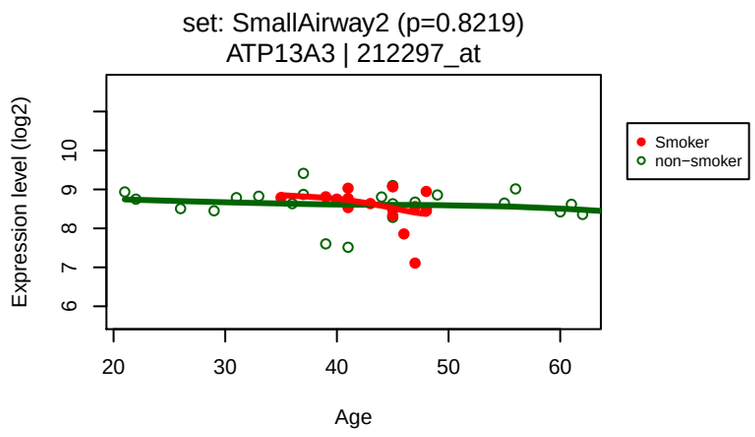
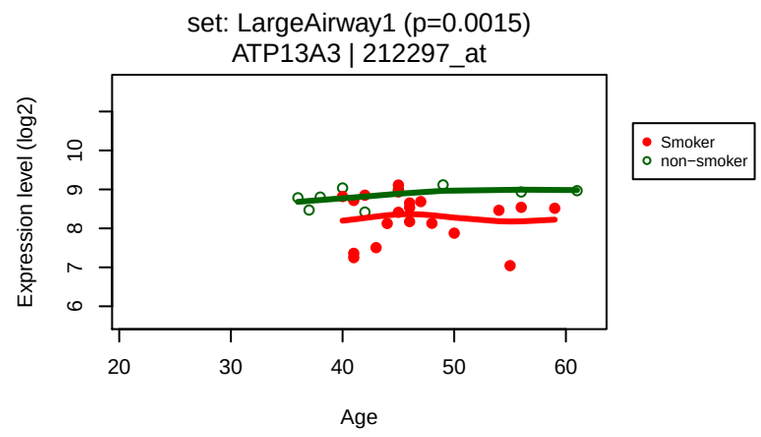

Gene expression of smokers and nonsmokers
gene: ATP13A3 | 212297_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.373
p=0.8219
p=0.0298
p=0.0015
p=0.0453
p=0.9516
p=0.0402
p=0.5719
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

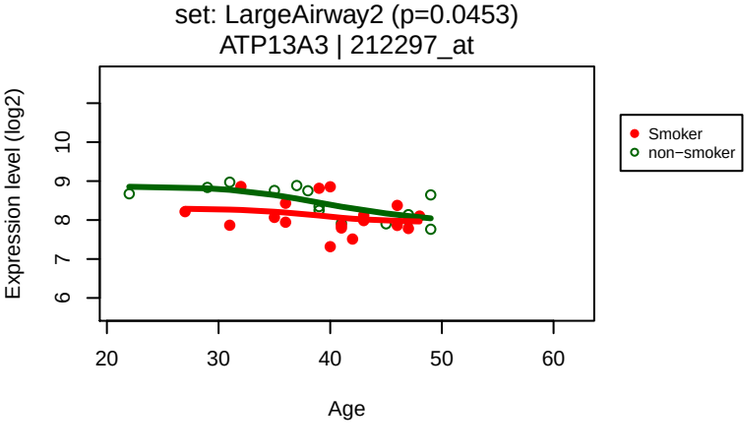

Gene expression ratio (smokers/nonsmokers)
gene: ATP13A3 | 212297_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.373
p=0.8219
p=0.0298
p=0.0015
p=0.0453
p=0.9516
p=0.0402
p=0.5719
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

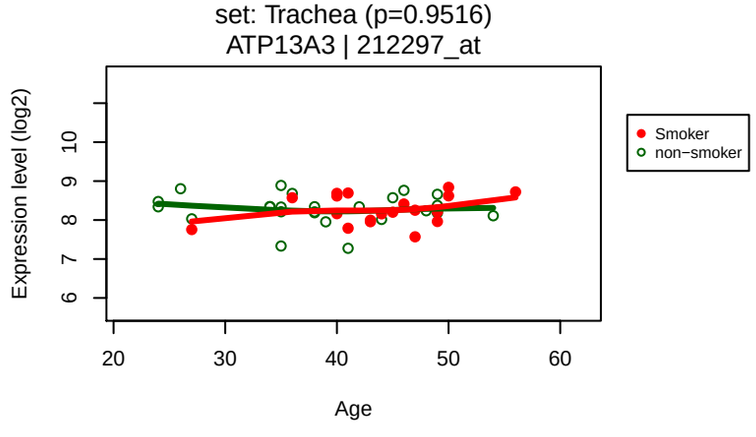

Gene expression ratio in Males
gene: ATP13A3 | 212297_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7416
p=0.018
p=0.5184
p=0.2576
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP13A3 | 212297_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3029
p=0.8416
p=0.397
p=0.3999
SmallAirway2
LargeAirway2
Trachea
Macrophage2

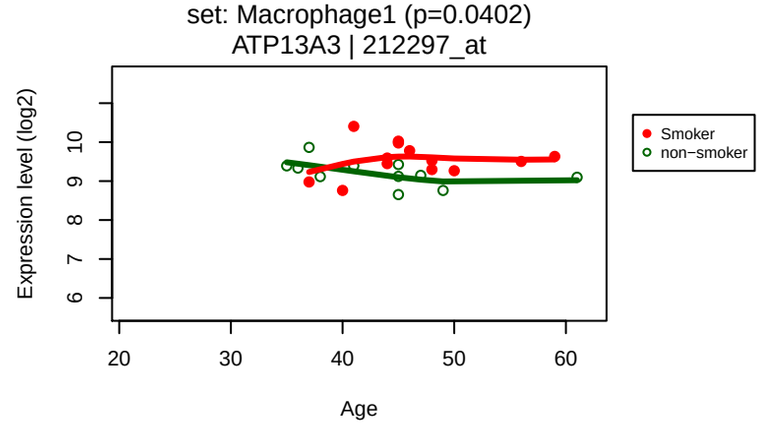

set: SmallAirway1 (p=0.373)
ATP13A3 | 212297_at
Expression level (log2)
Age
Smoker
non−smoker

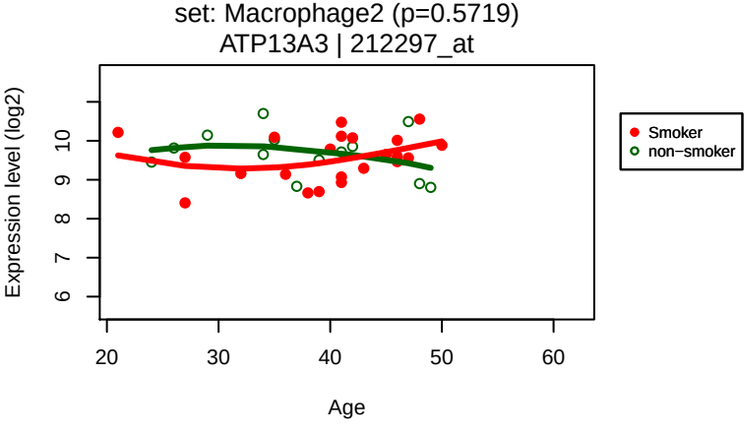

set: SmallAirway2 (p=0.8219)
ATP13A3 | 212297_at
Expression level (log2)
Age
Smoker
non−smoker

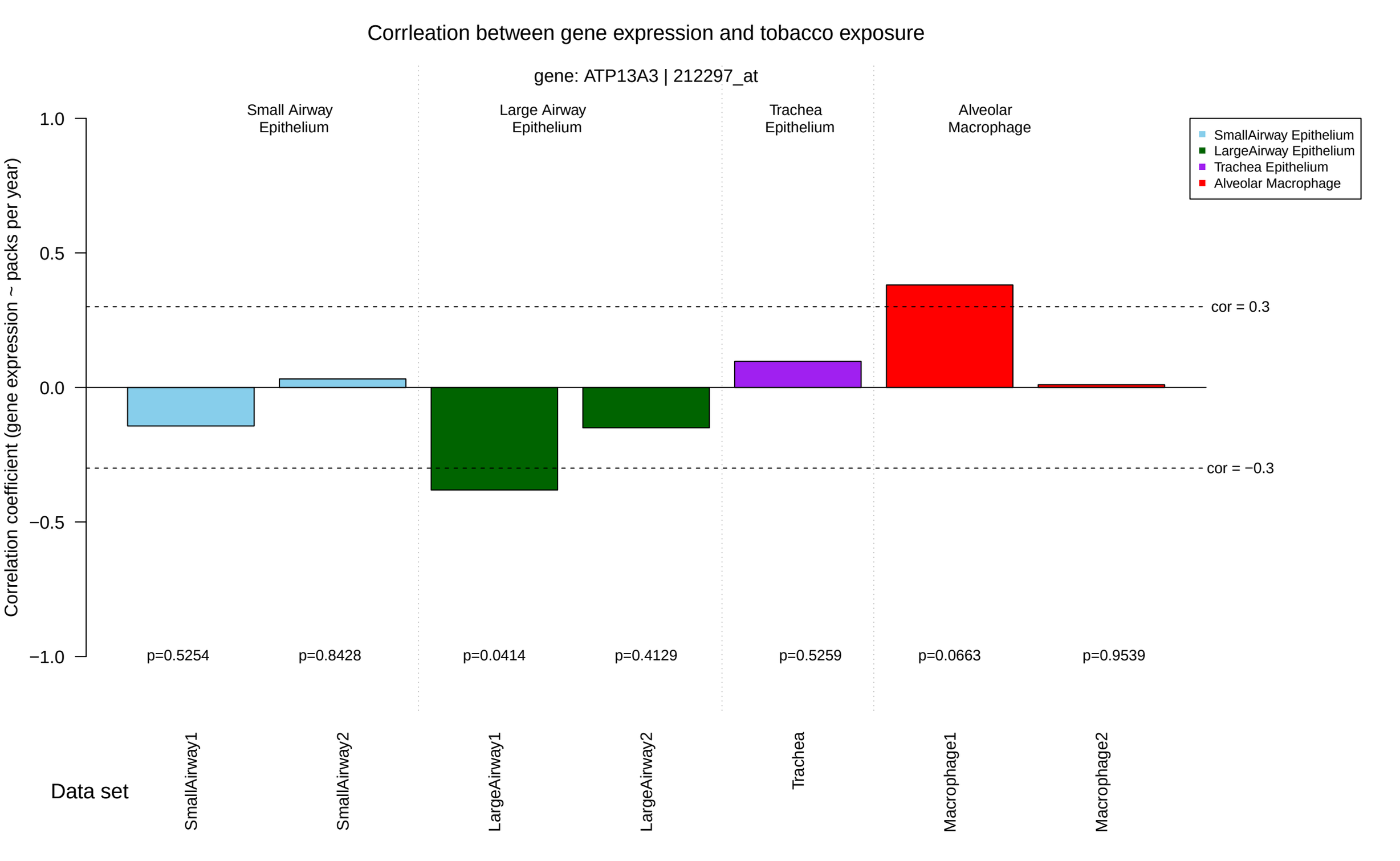

set: LargeAirway1 (p=0.0015)
ATP13A3 | 212297_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0453)
ATP13A3 | 212297_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9516)
ATP13A3 | 212297_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0402)
ATP13A3 | 212297_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5719)
ATP13A3 | 212297_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP13A3 | 212297_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5254
p=0.8428
p=0.0414
p=0.4129
p=0.5259
p=0.0663
p=0.9539
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2